

Unit 11, Lesson 3: Constant of Proportionality

Part 2

Benchmark

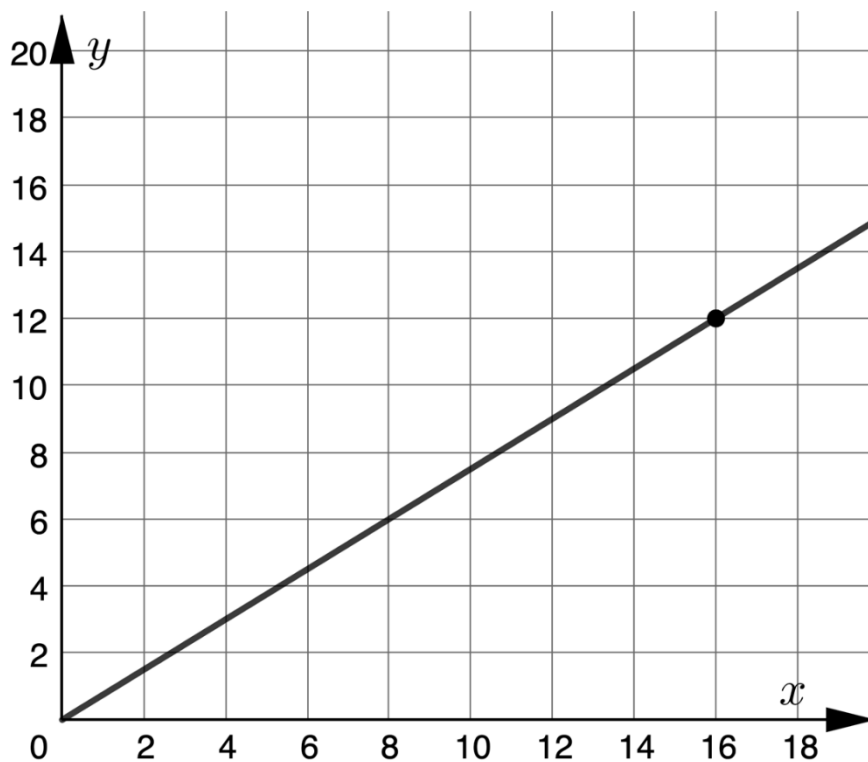
MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.

Learning Targets

- ☐ I can determine the constant of proportionality from a written description.
- ☐ I can relate the constant of proportionality to an equation representing a proportional relationship.

11.3.1 Warm-Up: Bell Work

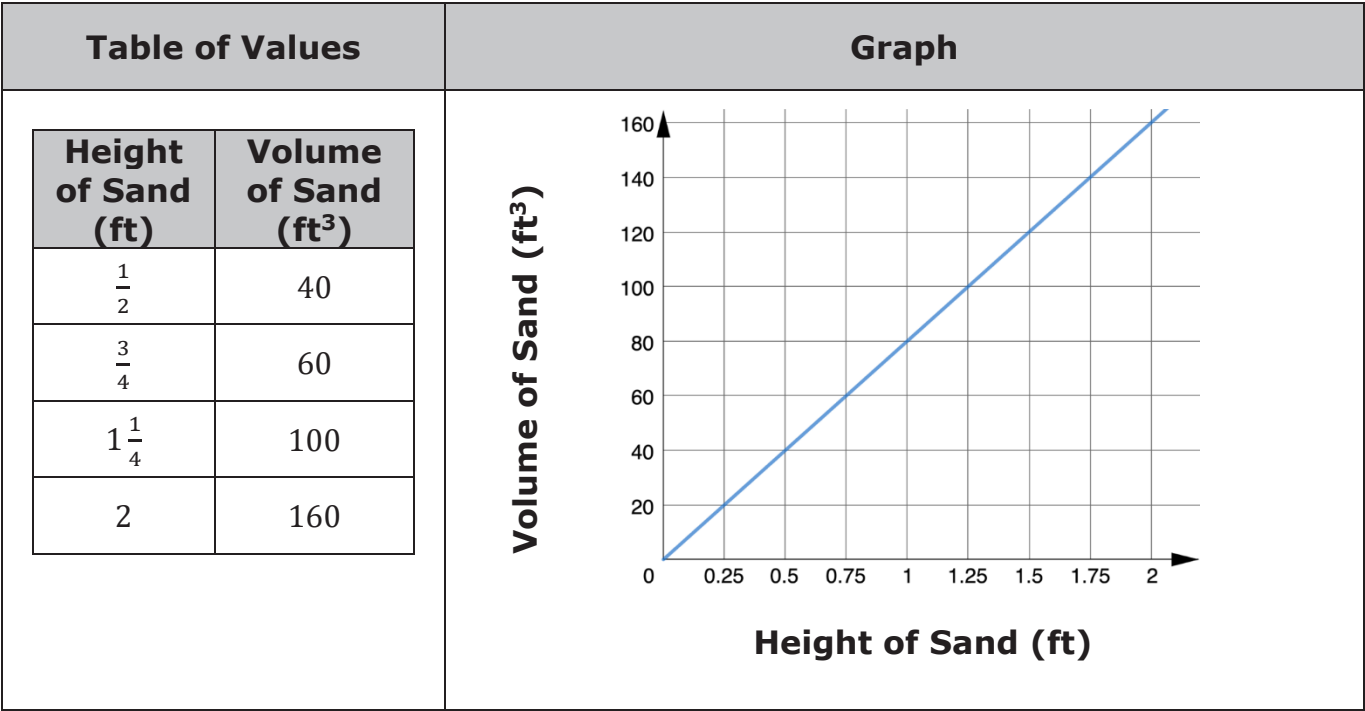
The graph represents a proportional relationship.



1. Describe a situation that could be represented by this graph.
2. Label the axes with the quantities in your situation.
3. There is a point marked at (16,12) on the graph. What does this point represent in your situation?

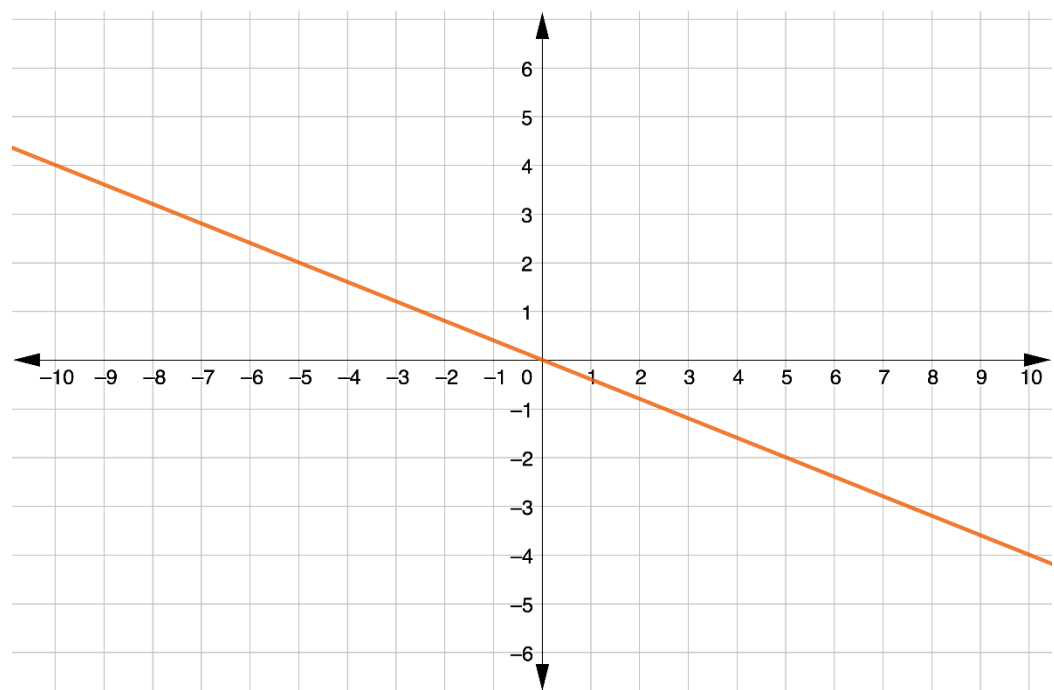
11.3.2 Guided Instruction: Relating the Constant of Proportionality to Written Descriptions

1. The volume of sand in a rectangular sandbox, y , changes proportionally with relationship to the height of the sand in the sandbox, x . The table of values and graph represent this relationship for a rectangular sandbox measuring 10 feet (ft) long and 8 ft wide with the capacity to fill to a height of 2 ft.



- a. Determine the constant of proportionality, $\frac{y}{x}$, using the table of values. Show your work.
- b. Explain how to determine the constant of proportionality using the graph.
- c. Describe the meaning of the constant of proportionality in this context.

2. The graph of a proportional relationship is shown.



a. Complete the table of values to represent the same relationship.

<i>x</i>	-5	-3	0	1	5	6.2
<i>y</i>						

- b. Verify the values in your table with your partner and resolve any differences, if necessary.
- c. What is the constant of proportionality, $\frac{y}{x}$, in this relationship?
- d. Verify the ratio of $\frac{y}{x}$ is constant for each coordinate pair represented in your table using a calculator. Make edits to your work as needed.

- e. Discuss with your partner how the constant of proportionality can be used to determine the y -values corresponding to specific x -values in the table in cases where it is difficult to determine from looking at the graph.
- f. This proportional relationship can be used to represent the relationship between the original price, x , and the change in price due to a store-wide sale, y . Complete the statement to interpret the constant of proportionality found in Part C in this context.

The constant of proportionality is _____, so the

change in price due to the sale is a _____%

- ☐ reduction
 - ☐ mark up

from the original price. This means

the store is having a _____% off sale.

- g. Explain which part(s) of the graph representing this proportional relationship would not make sense mathematically for this real-world context.

11.3.3 Guided Practice: Determining the Constant of Proportionality from a Written Description

1. Kyla is training for a half-marathon. She sets her treadmill to a constant speed to build up her endurance on longer runs. The relationship between the time she runs, x , and the distance she runs, y , is proportional. She runs 10.5 miles in 1.75 hours at this pace.

a. Determine the constant of proportionality, $\frac{y}{x}$.

- b. Complete the statement to interpret the constant of proportionality.

At this rate, Kyla runs _____

☐ miles

☐ hours

 for every

☐ 1 mile

☐ 1 hour

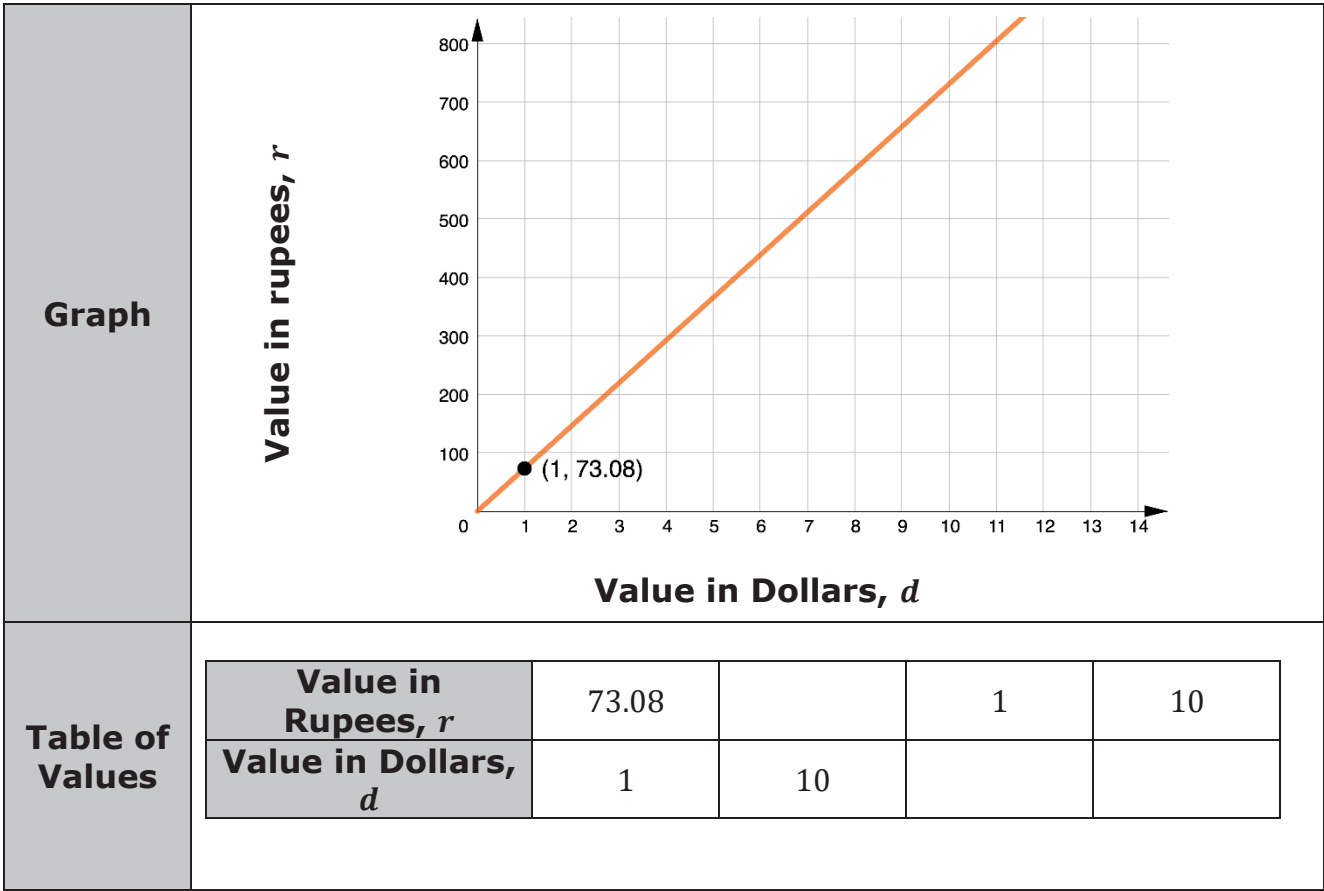
 she trains on the treadmill.



In a real-world context, the **constant of proportionality** is the amount one quantity increases or decreases each time the other quantity increases by 1 unit.

11.3.4 Collaboration: Equations Representing Proportional Relationships

On January 14, 2021, 1 U.S. dollar was worth 73.08 Indian rupees. This is a written description of a proportional relationship. Below are two other representations of the same relationship.



Work with your partner to complete the following.

1.
- Complete the table of values by filling in the missing values. Round values to the nearest thousandth, where necessary.

Proportional relationships can also be represented by equations.

2.
- Use the quantities determined in the table of values to complete each equation that represents this relationship.

$d = \boxed{} \cdot r$

or

$r = \boxed{} \cdot d$

3. What is the constant of proportionality, $\frac{d}{r}$, in this context?

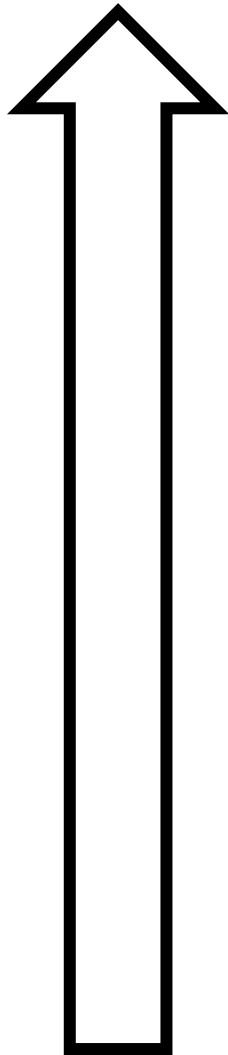
4. What is the constant of proportionality, $\frac{r}{d}$, in this context?

5. Discuss with your partner how the constants of proportionality relate to the equations of the proportional relationship.

11.3.5 Wrap-Up: Reflection

1. Color in the arrow up to the statement that best describes your current understanding of the benchmark below.

Determine the constant of proportionality within a mathematical or real-world context given one of these forms: a table, graph, or written description of a proportional relationship.



I'm so confident, I could explain how to determine the constant of proportionality for proportional relationships to someone else!

I can get to the right answer, but I don't understand well enough to explain how to determine the constant of proportionality yet.

I understand how to find the constant of proportionality for some representations of proportional relationships, but I don't understand how to for all forms yet.

I have been trying hard, but I am not yet able to determine the constant of proportionality for proportional relationships.

I do not understand how to determine the constant of proportionality for proportional relationships yet. There are things I could do to be a better learner next lesson.

